Ritesh Sur Chowdhury

riteshsurchowdhury2001@gmail.com | +91-6290769461

Portfolio | linkedin | Google Scholar | Researchgate | ORCID

About Me

I am pursuing an MS by Research in Data Science and AI at IIT Bombay under Prof. Subhasis Chaudhuri and Prof. Biplab Banerjee, and am currently a Visiting Student at MBZUAI under Prof. Junpei Komiyama.
Research Interests: Deep Learning, CV, NLP, LLMs, VLMs, Domain Generalization

Education

- **Indian Institute of Technology (IIT), Bombay** July 2024 - Present
  MS by Research in Centre for Machine Intelligence and Data Science (CMInDS), **GPA: 9.66/10**
- **Jadavpur University** August 2018 - July 2022
  B.E. in Electronics and Telecommunication Engineering (ETCE), **GPA: 9.61/10**

Experience

- **MBZUAI – Masdar City, UAE** August 2026 – Present
  – **Position:** Visiting Student under Prof. Junpei Komiyama,
  Research Topic: On Policy Self Distillation in LLMs
- **IBM Research – Bangalore, India** May 2026 – August 2026
  – **Position:** Research Intern , **Team:** Data and AI
  – **Research Topic:** KG-Eval: A Unified Framework for Structural and Utility-Based Knowledge Graph Assessment (Submitted in ARR October 2026)
- **Qualcomm – Hyderabad, India** July 2022 – July 2024
  – **Position:** Engineer (December 2023 – July 2024), Associate Engineer (July 2022 – December 2023),
  Team: FSM System Test (FST) , **Job Description:** System Testing and Automation Development
  – **Awards:** Impact Award

Graduate research works

* indicates Joint first authorship.

- Domain-Robust PointMamba through Structured Cosine Perturbation
  – Accepted in WACV 2027
  – Ritesh Sur Chowdhury*, Mohammed Nasir*, Muhammad Haris Khan, Subhasis Chaudhuri, Biplab Banerjee.
  – **Problem Statement:** We designed a Mamba-based spectral model for domain generalization in 3D point clouds, addressing the single-source setting where models must generalize to unseen target domains.
- Scaled Autoregressive Projection Steering (SAPS): Token-level Steering to Induce Sarcasm and Humblebrag in Text
  – Accepted in EMNLP 2026 (Findings)
  – Sharath Naganna*, Ritesh Sur Chowdhury*, Dipankar Srirag, Aditya Joshi
  – **Problem Statement:** This work investigates the induction of incongruity-based figurative language (sarcasm and humblebrag) in large language models using a token-level steering approach.
- Pedestrian Trajectory Prediction for autonomous vehicles using dashcam views.
  – Supervised by Prof. Biplab Banerjee, and collaboration with Honda R&D Japan.
  – **Problem Statement:** Existing methods use the same models for both bird's-eye-view (BEV) and dashcam views. We focus on evaluating and optimizing dashcam-specific modelling for autonomous driving.
- The Underlying Mechanistic Connection between Zero-Shot and In-Context Learning
  – Supervised by Prof. Aditya Joshi and Prof. Ashutosh Modi.
  – **Problem Statement:** This work investigates the mechanistic relationship between instruction-only zero-shot and demonstration-only ICL circuits, examining how ICL induction builds upon and extends zero-shot inference.

Undergraduate Publications

* indicates Joint first authorship.

- S. Bose*, **R. S. Chowdhury***, D. Pal, S. Bose, B. Banerjee, and S. Chaudhuri, Multiscale probability map guided index pooling with attention-based learning for road and building segmentation. ISPRS Journal of Photogrammetry and Remote Sensing. 2023 Dec 1;206:132-48.
- **R. S. Chowdhury***, S. Bose*, S. Ghosh, and A. Konar, "Attention induced dual convolutional-capsule network (AIDC-CN): A deep learning framework for motor imagery classification," *Computers in Biology and Medicine*, vol. 183, p. 109260, 2024.
- S. Bose*, **R. S. Chowdhury***, R. Das, and U. Maulik, "Dense dilated deep multiscale supervised U-network for biomedical image segmentation," *Computers in Biology and Medicine*, vol. 143, p. 105274, 2022.
- R. Das*, S. Bose*, **R. S. Chowdhury**, and U. Maulik, "Dense dilated multi-scale supervised attention-guided network for histopathology image segmentation," *Computers in Biology and Medicine*, p. 107182, 2023.
- S. Ghosh, S. Bose, **R. S. Chowdhury**, A. Konar, and A. K. Nagar, "Decoding the creative ability of subjects from aesthetic quality assessment using dual convolution induced capsule network," in *Proc. IEEE Symp. Series on Computational Intelligence (SSCI)*, pp. 1319–1326, 2022.
- **R. S. Chowdhury**, J. Jana, S. Tripathi, and J. Bhaumik, "Improved DWT and IDWT architectures for image compression," *Microprocessors and Microsystems*, p. 104990, 2023.
- **R. S. Chowdhury**, J. K. Sadhu, C. Thakur, and S. Chattopadhyay, "Performance analysis and optimization of a hybrid TSR–PSR protocol for AF, DF and hybrid AF–DF relaying under Weibull fading," *Telecommunication Systems*, vol. 82, no. 1, pp. 61–90, 2023.
- J. Jana, **R. S. Chowdhury**, S. Tripathi, and J. Bhaumik, "FPGA implementation of compact and low-power multiplierless architectures for DWT and IDWT," *Journal of Real-Time Image Processing*, vol. 21, no. 1, p. 19, 2024.
- J. Jana, S. Tripathi, A. Bhattacharya, **R. S. Chowdhury**, and J. Bhaumik, "An area efficient VLSI architecture for 1-D and 2-D discrete wavelet transform (DWT) and inverse discrete wavelet transform (IDWT)," in *Proc. Devices for Integrated Circuit (DevIC)*, pp. 378–382, 2021.
- J. Jana, S. Tripathi, A. Bhattacharya, **R. S. Chowdhury**, A. Karmakar, D. Ranjan, and J. Bhaumik, "An Android-application-controlled car for human safety against COVID-19," in *Internet of Things and Its Applications: Select Proceedings of ICIA 2020*, pp. 3–12, Springer, 2022.
- J. Jana, S. Tripathi, A. Bhattacharya, **R. S. Chowdhury**, D. Ranjan, and J. Bhaumik, "A cost-effective tracking and health monitoring system for suspected COVID-19 patient in quarantine," in *Microelectronics, Circuits and Systems: Select Proceedings of Micro2021*, pp. 569–579, Springer, 2023.

Journal Reviews

- **Journals:** Information Fusion, International Journal of Computer Vision, ISPRS Journal of Photogrammetry and Remote Sensing.

Scholastic Achievements & Competitions

- **Heidelberg-Laureate Forum (HLF)** 2026
  Invited to the annual HLF program as one of 202 selected young researchers.
- **Graduate Aptitude Test in Engineering (GATE)** 2022
  Electronics and Communication Engineering, **All India Rank:** 59.
- **FIFS Gameathon** 2025
  1st position in Gameathon1.0 (ICC Champions League), **2nd position** in Gameathon2.0 (IPL)
  Challenge: Build an ML model to select optimal team of 11 players (Aggregated for a live tournament).
- **Smart India Hackathon (SIH)**: Selected for national level (IIT Bombay). 2025
  Problem Statement: Optical-Guided Super-Resolution for Thermal IR Imagery